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**2nd International Symposium on Innovations in Computer Vision and Applications
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**Multimodal Geometric Deep Learning for Urban Commercial Potential Segmentation
using Open Geospatial Data**

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INTRODUCTION



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Context

- Vietnam's Coffee & Tea sector is expanding rapidly and highly competitive
- Site selection is critical to long-term F&B success
- Owners need reliable tools to evaluate potential locations

Our Approach

- Use only open geospatial data: Google Maps reviews, OpenStreetMap POIs, VIIRS Nighttime Light
- Build proxy labels to replace missing revenue data
- Classify every 250m × 250m grid cell into Low / Medium / High potential across Hanoi, Da Nang, HCMC

Problem

- Data-driven models rely on proprietary data: revenue, transactions, mobility
- Such data is unavailable in fragmented markets, so independent operators are left out
- Open data sources exist but are rarely combined for fine-grained site evaluation

Research Direction	Main Data Sources	Remarks
Traditional geospatial evaluation	Demographic, footfall, revenue data	Interpretable but depends on proprietary business data
Open-data POI-based analysis	OSM POI networks, spatial accessibility	Works without sales records but weak interaction modeling
Graph-based modeling	POI graphs, check-ins, mobility data	Strong spatial interaction modeling but requires O2O data
NTL-based economic modeling	VIIRS NTL satellite imagery	Good macroeconomic proxy but coarse spatial resolution
Multimodal learning	POI, reviews, interaction logs, spatial tensors	Strong multimodal fusion but requires dense behavioral data

The Gaps: Existing methods either **depend on proprietary data** or, when using open data, **lack multimodal spatial learning**, leaving **no accessible framework for fine-grained site evaluation** in data-scarce markets.



MAIN CONTRIBUTIONS



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1

Open-Data Proxy Labeling

A labeling method that generates 3-level site potential without proprietary revenue data, by combining: Google Maps ratings & review sentiment, OSM POI networks, VIIRS Nighttime Light

2

Multimodal Deep Learning Model

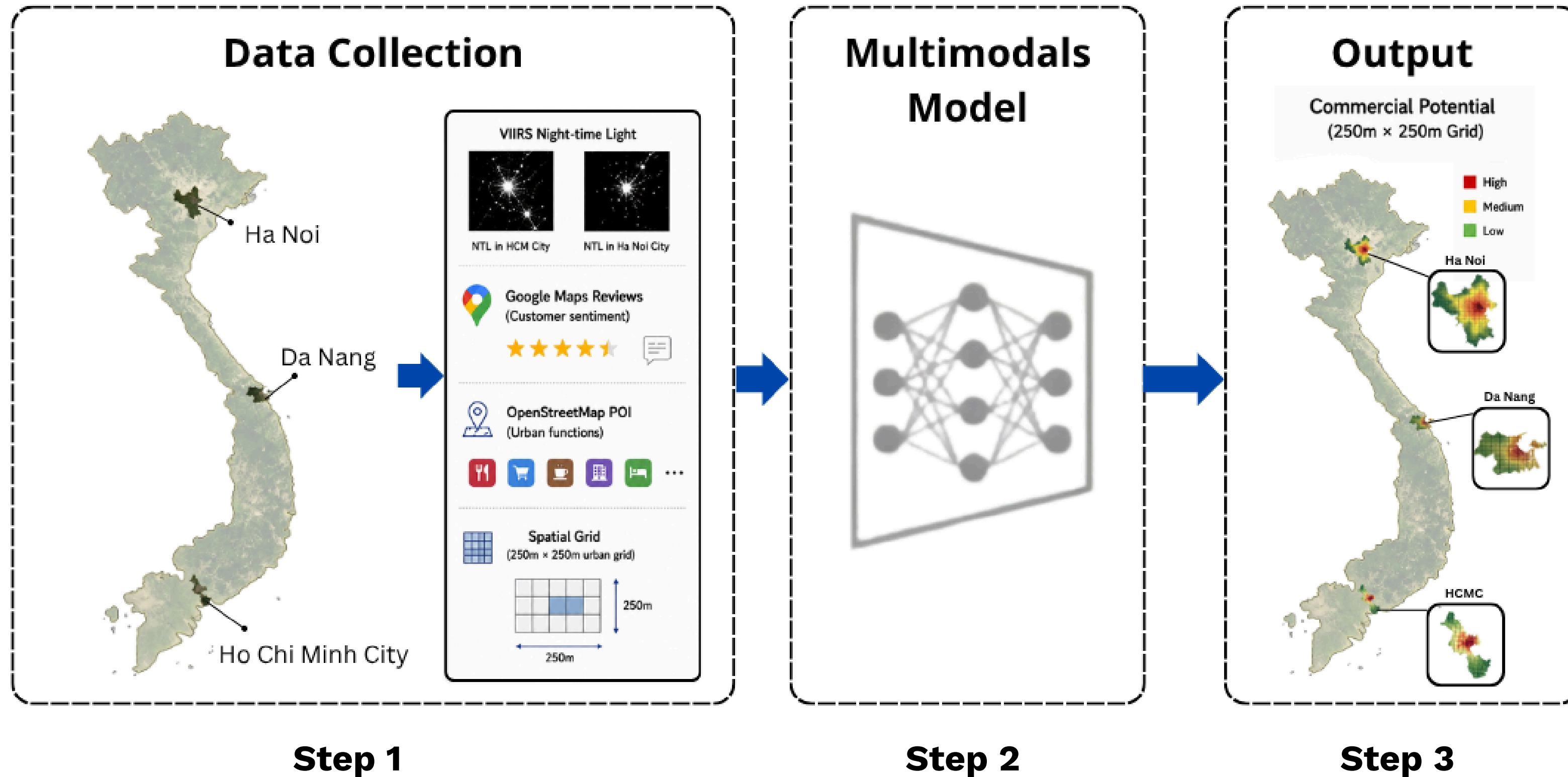
An adapted Wide & Deep + Cross Network that jointly learns from: Tabular features of each cell and 11×11×11 spatial tensors of the surrounding neighborhood, via channel-spatial attention

3

Fine-grained Urban Benchmark

- A 250m × 250m grid dataset for Vietnam's Coffee & Tea market:
- 73,974 grid cells across Hanoi, Da Nang, and HCMC
 - 8,647 Coffee & Tea venues, 51,727 POIs

PIPELINE OVERVIEW





DATA COLLECTION



Data from Google Maps

- **Identification & spatial data:** Shop name, address, precise coordinates (latitude, longitude), category.
- **Rating data:**
 - **Rating:** Reflects service/product quality.
 - **Reviews Count:** Reflects popularity, customer traffic, and brand recognition.

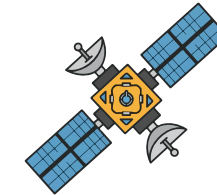


Data from OpenStreetMap

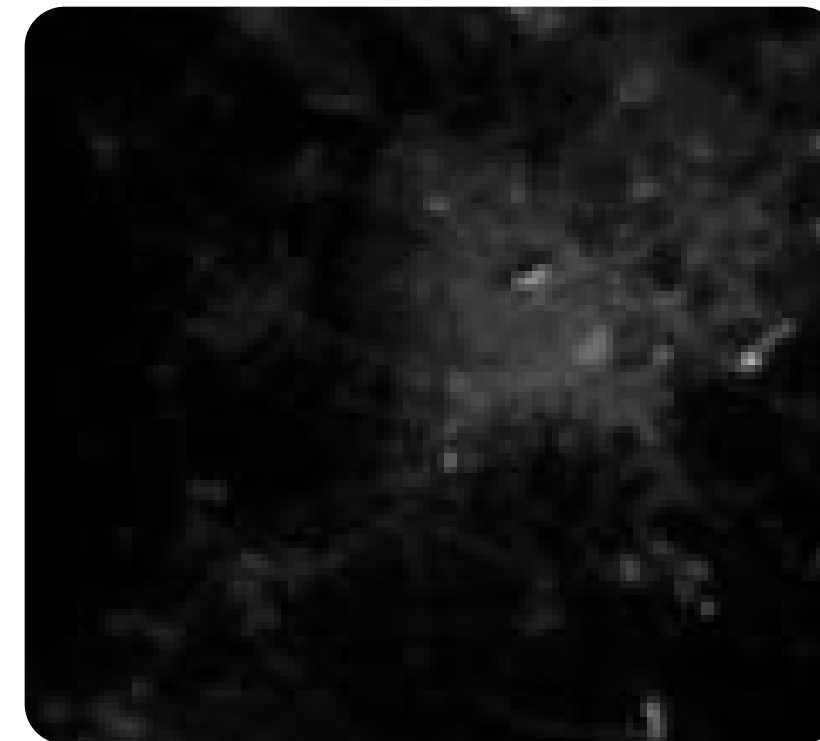
- **Features:** POI data categorized into 8 groups
- **Objective:** Capture micro-level urban function via POI counts, density and entropy.



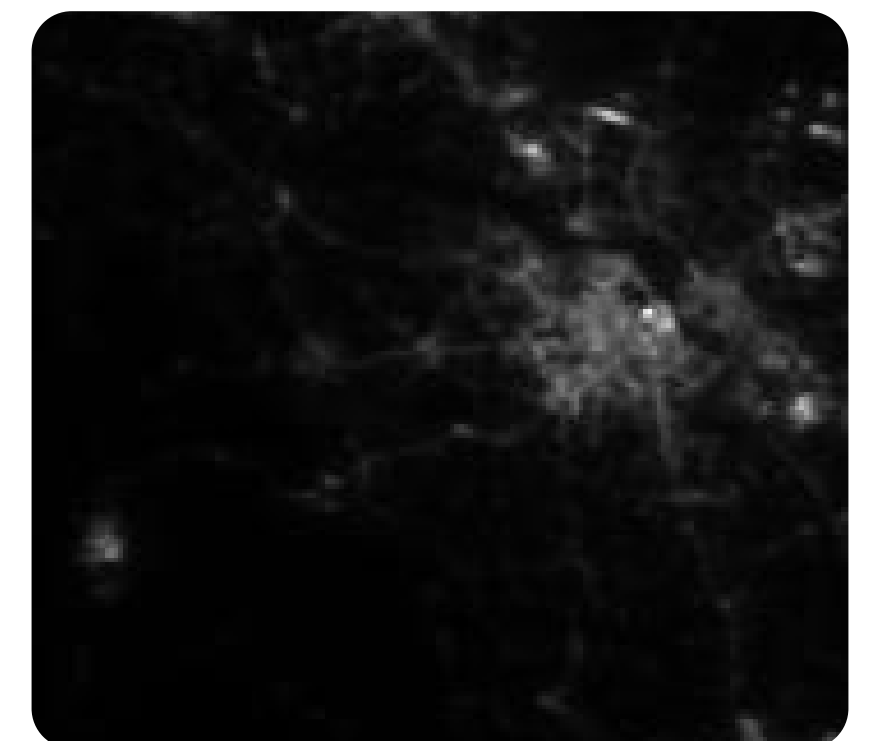
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Nighttime Light (NTL) Data



NTL over HCMC



NTL over Hanoi

- **Features:** NTL_log and NTL_neighbor
- **Objective:** Objectively measure economic development, commercial vibrancy, and nighttime activity intensity at the macro level.



FEATURE ENGINEERING



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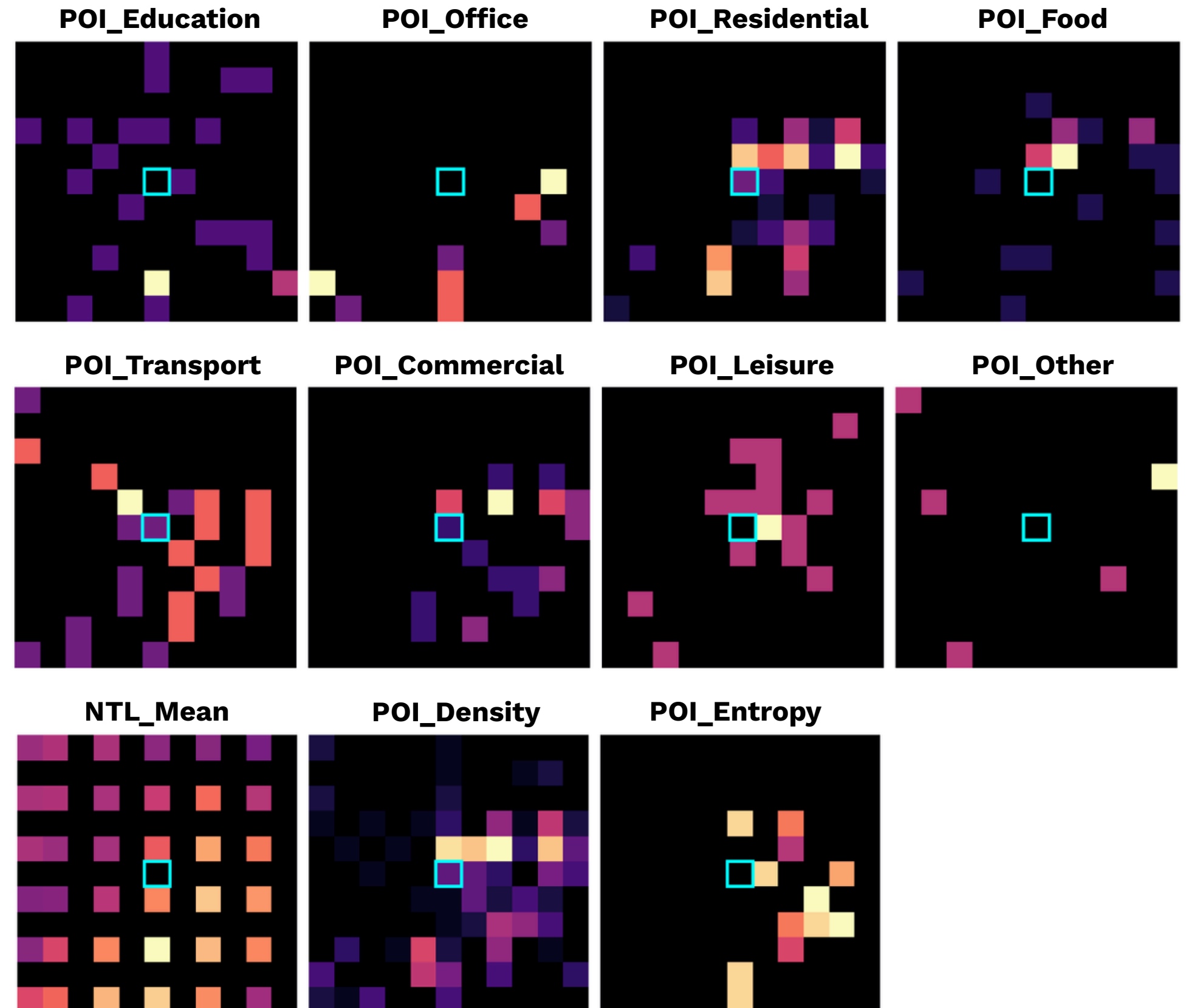
Spatial Feature Matrix

Method

- Centering on the target grid cell, extract the neighborhood within a 5-cell radius (creating an 11x11 matrix).
- A depth of 11 channels: 8 POI categories + NTL mean, POI density, POI entropy

Objective

- Preserve the geographical structure (distance, direction) => Transform neighborhood data into a "multi-channel image" to allow the model to learn the micro-context.
- Standardized input ready for the Spatial Embedding layer.





LABEL ENGINEERING

Core idea

Indirectly measure "Location potential" through three factors:

- **Popularity:** Number of establishments and total number of reviews (Review Count).
- **Service quality:** Normalized score (Bayesian Average Rating) and Sentiment Score.
- **Environment:** Level of economic vibrancy derived from remote sensing (NTL).

Mathematical Formulation

$$V_{raw} = \ln(1 + N_{cafe}) \times \ln(1 + N_{review}) \times \left(\frac{\frac{R_{bayes}}{5} + S_{nlp}}{2} \right) \times E_{neighbor}$$

Where:

$$E_{neighbor} = 0.5 + \frac{NTL_{neighbor}}{NTL_{neighbor,max}}$$



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Classification logic

📌 **Case 1:** Grid cell with **existing shops**

$V_{raw} > Median \Rightarrow$ Class 2: High potential

$V_{raw} \leq Median \Rightarrow$ Class 1: Medium potential

📌 **Case 2:** **Empty** grid cell

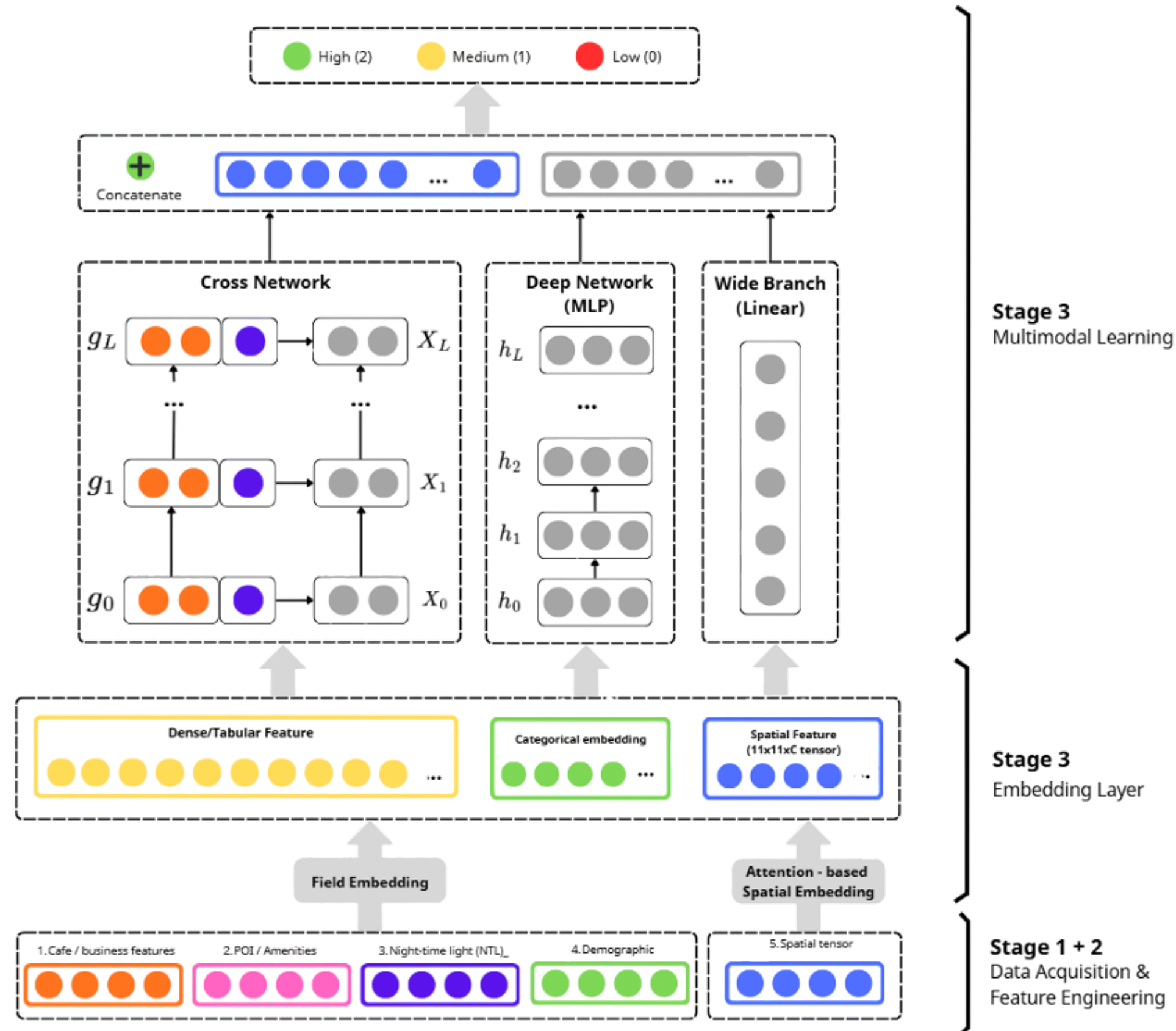
$NTL_{neighbor} \geq Threshold$

$\Rightarrow$ Class 1: Latent / untapped potential

$NTL_{neighbor} < Threshold$

$\Rightarrow$ Class 0: Low potential

PROPOSED OVERVIEW



Output

3-class label (Low/Medium/High) → potential heatmap

Multimodal Learning

Wide + Cross + Deep → concatenate → softmax

Embedding Layer

- Field Embedding (tabular/categorical)
- Attention-based Spatial Embedding (channel + spatial)

Input Features

13-dim tabular features and 11×11×11 spatial tensor

Table 2. Performance comparison on the test set.

Model	Accuracy	F1 Macro	F1 Weighted
Logistic Regression	0.9165	0.7073	0.9219
Bagging (LR base)	0.9303	0.7535	0.9337
DNN Baseline	0.8699	0.6215	0.8865
Wide & Deep	0.8667	0.6167	0.8844
DCN	0.8700	0.6364	0.8873
Proposed Model	0.9449	0.7613	0.9489

Table 3. Ablation study results on the held-out test set. Δ denotes change in Macro F1 relative to the full model.

Model	Accuracy	F1 Macro	Δ F1 Macro	Training time (s)
Full model	0.9449	0.7613	-	73.3
Only Tabular data	0.8744	0.6400	-0.1213	70.3
Only Spatial data	0.9230	0.6897	-0.0716	46.8
w/o Channel attention	0.9346	0.7309	-0.0304	43.5
w/o Spatial attention	0.8932	0.6494	-0.1119	68.7
w/o Cross Network	0.9467	0.7563	-0.0050	66.1
w/o Deep Network	0.9305	0.7223	-0.0390	45.9
w/o Wide Branch	0.9351	0.7232	-0.0381	47.9



CONCLUSION

Contribution

An open-source geospatial proxy-label framework with a multimodal architecture (spatial-channel attention + Cross/Deep/Wide branches) for Coffee & Tea site potential in data-scarce markets.

Key results

Macro F1 = 0.7613 across 14,797 grid cells (3 cities), outperforming all baselines, with 0.84 recall on high-potential sites; spatial attention is the most critical component.



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Limitations

Proxy labels depend on existing business density, are unvalidated against real revenue, and are evaluated only within Vietnam.

Future work

Incorporate rent/mobility data, multilingual sentiment analysis, longitudinal analysis, and cross-country validation.



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THANK YOU FOR LISTENING!

Question & Discussion:

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